

YASH RAWAL

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PROFESSIONAL SUMMARY

Applied AI & Machine Learning Engineer serving a **1M+ user scale** by architecting enterprise agentic platforms, fine-tuning domain embedding models (**E5-large-v2**) on NVIDIA GPUs, and building automated evaluation control planes. Specialized in **Python**, high-throughput LLM serving internals (vLLM contributor), Java, Spring Boot, SAP BTP, workflow orchestration (LangGraph, n8n), and cloud-native infrastructure (Azure, AWS, Docker, Kubernetes). Active open-source contributor to Spring AI and vLLM.

TECHNICAL SKILLS

Languages & Backend: Python, SQL, FastAPI, PostgreSQL, Java, Spring Boot, Spring AI, SAP BTP, Oracle, REST APIs, WebSockets, AsyncIO, Linux
AI & ML Engineering: Agentic AI, Model Context Protocol (MCP), Multi-Agent Systems, Tool RAG, Healthcare NLP, E5-large-v2 Fine-Tuning, vLLM Serving
AI & Workflow Frameworks: LangChain, LangGraph, n8n, vLLM, PyTorch, Hugging Face Transformers, SAP Joule 2.0, LlamaIndex, GROBID
Cloud & Infrastructure: Azure, AWS, Docker, Kubernetes, CI/CD (GitHub Actions, Jenkins, Git, Gerrit), Redis, PGVector
Testing, QA & Security: PyTest, LLM-as-a-Judge, Prompt Lab, OpenAI Red Teaming, PII Sanitization, Unit/Integration/Regression Testing

WORK EXPERIENCE

Bigscal Technologies Pvt. Ltd.

AI/ML Engineer Oct 2024 - Present
Surat, India

- Deployed a production-scale agentic platform and multi-agent systems serving **1M+ active ERP users** across 120+ enterprise tenants on cloud infrastructure (Azure, AWS, SAP BTP).
- Architected multi-agent routing with **LangGraph, n8n & SAP BTP**, implementing hybrid semantic-lexical tool discovery (vector + BM25F/RRF) across 650+ dynamic read/write and generation MCP capabilities using Python and Java/Spring Boot.
- Designed a capability contract layer to execute nested profiles inside a single ReAct loop, reducing execution overhead by **3x** and cutting token consumption by **70%**, slashing enterprise inference costs from \$150 to \$30 per 1K multi-agent runs.
- Fine-tuned and served **E5-large-v2** embedding models on NVIDIA GPUs using PyTorch, Unsloth, and contrastive learning for high-precision role mapping and semantic retrieval.
- Implemented a 4-level hierarchical RAG pipeline in Python indexing 10,000+ technical documents using GROBID, pdfplumber, and PostgreSQL/PGVector, achieving a **90%+ Recall@10** retrieval rate.
- Designed the **Phantom Token** security pattern to pass authenticated user credentials securely between custom agents and Model Context Protocol (MCP) servers without exposing raw JWT payloads.
- Developed the Prompt Lab evaluation control plane in Python to collect reasoning trajectories and human preference annotations, establishing a scalable **Human Feedback (RLHF) Pipeline**.

Bigscal Technologies Pvt. Ltd.

AI/ML Engineer Intern Apr 2024 - Sep 2024
Surat, India

- Engineered clinical summarization pipelines in Python for 116K+ medical records with 4-bit quantized Llama 3, and designed real-time WebSocket audio streaming for Indic ASR with **NeMo RNN-T and CTC decoding**.
- Researched audio digital signal processing (STFT, Mel-spectrograms, MFCC feature extraction via librosa/torchaudio) and CNN audio architectures (VGGish, ResNet), compiling custom Indic LJSpeech-format datasets.

OPEN SOURCE & ENGINEERING PROJECTS

Spring AI Framework (spring-projects/spring-ai) Open Source
Merged **PR #5711** in Spring AI restoring missing `reasoningContent` metadata fields for DeepSeek-R1 and Qwen reasoning models, and contributed Amazon Bedrock Converse API prompt-cache time-to-live (TTL) configuration parameters.
Java, Spring Boot, Spring AI, Amazon Bedrock, REST APIs

vLLM High-Performance Inference Engine (vllm-project/vllm) Open Source
Merged **PR #55475** fixing structural XML tool parser drops on `v1/chat/completions` and Pydantic parameter indexing (64/64 Qwen tests passed); engineered `torchcodec` multimodal decoder guards (Issue #54097).
Python, PyTorch, vLLM, Pydantic, AsyncIO, PyTest

Enterprise Hierarchical Document Intelligence & RAG Platform Project
Built an **immutable document versioning system** and four-level hierarchical knowledge ingestion pipeline (paragraphs, pages, chapters, headings) using GROBID and pdfplumber; implemented paragraph-level signature hashing in `pgvector` achieving a **90%+ Recall@10** retrieval rate.
Python, PostgreSQL, pgvector, GROBID, pdfplumber, RAG

E5-large-v2 Job & Skills Embedding Fine-Tuning Model Training
Fine-tuned **E5-large-v2** (335M parameters) on Lightcast labor-market data using PyTorch and triplet loss (anchor + positive + hard negative), raising semantic retrieval relevance by **18%** over OpenAI text-embedding-3-small.
Python, PyTorch, Hugging Face Transformers, Triplet Loss, NVIDIA GPUs

OpenAI Agentic Red-Teaming & Security Research Kaggle Competition & Research
Engineered an **agentic red-teaming framework** inspired by OpenAI red-teaming benchmarks and Kaggle adversarial competitions to automatically mutate prompt payloads, challenge observer guardrails, and detect LLM security vulnerabilities.
Python, PyTorch, LLM-as-a-Judge, Adversarial Red Teaming, Jailbreak Datasets

EDUCATION

Parul University 2022 - 2024
Master of Computer Applications (Artificial Intelligence & Machine Learning)

CERTIFICATIONS

Microsoft Certified: Azure AI Fundamentals (AI-900) **Microsoft**
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